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03. PyTorch Computer Vision Exercises

The following is a collection of exercises based on computer vision fundamentals in PyTorch.

They're a bunch of fun.

You're going to get to write plenty of code!

Resources

1. These exercises are based on notebook 03 of the Learn PyTorch for Deep Learning course.
2. See a live walkthrough of the solutions (errors and all) on YouTube.

- **Note:** Going through these exercises took me just over 3 hours of solid coding, so you should expect around the same.

3. See other solutions on the course GitHub.

```
# Check for GPU
!nvidia-smi
```

```
Tue Dec 30 15:35:26 2025
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 550.54.15              Driver Version: 550.54.15      CUDA V
|-----------------------------------------+------------------------+-----
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Vola
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-
|                                         |                        |
|=========================================+========================+=====
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |
| N/A   74C    P8             23W /   70W |       0MiB /  15360MiB |
|                                         |                        |
+-----------------------------------------+------------------------+-----

+-----------------------------------------------------------------------------+
| Processes:
|  GPU   GI   CI        PID   Type   Process name
|        ID   ID
|=============================================================================
|  No running processes found
+-----------------------------------------------------------------------------+
```

```
# Import torch
import torch

# Exercises require PyTorch > 1.10.0
print(torch.__version__)

# TODO: Setup device agnostic code
```

```
2.9.0+cu126
```

1. What are 3 areas in industry where computer vision is currently being used?

1. Self-driving cars, such as Tesla using computer vision to percieve what's happening on the road. See Tesla AI day for more - https://youtu.be/j0z4FweCy4M
2. Healthcare imaging, such as using computer vision to help interpret X-rays. Google also uses computer vision for detecting polyps in the intenstines - https://ai.googleblog.com/2021/08/improved-detection-of-elusive-polyps.html
3. Security, computer vision can be used to detect whether someone is invading your home or not - https://store.google.com/au/product/nest_cam_battery?hl=en-GB

2. What is overfitting in machine learning?

Overfitting is like memorizing for a test but then you can't answer a question that's slightly different.

In other words, if a model is overfitting, it's learning the training data *too well* and these patterns don't generalize to unseen data.

3. Ways to prevent overfitting in machine learning

See this article for some ideas: https://elitedatascience.com/overfitting-in-machine-learning

3 ways to prevent overfitting:

1. Regularization techniques - You could use dropout on your neural networks, dropout involves randomly removing neurons in different layers so that the remaining neurons hopefully learn more robust weights/patterns.

2. Use a different model - maybe the model you're using for a specific problem is too complicated, as in, it's learning the data too well because it has so many layers. You could remove some layers to simplify your model. Or you could pick a totally different model altogether, one that may be more suited to your particular problem. Or... you could also use transfer learning (taking the patterns from one model and applying them to your own problem).
3. Reduce noise in data/cleanup dataset/introduce data augmentation techniques - If the model is learning the data too well, it might be just memorizing the data, including the noise. One option would be to remove the noise/clean up the dataset or if this doesn't, you can introduce artificial noise through the use of data augmentation to artificially increase the diversity of your training dataset.

4. CNN Explainer website.

`The convolutional neuron performs an elementwise dot product with a unique kernel and the output of the previous layer's corresponding neuron. This will yield as many intermediate results as there are unique kernels. The convolutional neuron is the result of all of the intermediate results summed together with the learned bias.`

5. Load the `torchvision.datasets.MNIST()` train and test datasets.

```
import torchvision
from torchvision import datasets

from torchvision import transforms
```

```
# Get the MNIST train dataset
train_data = datasets.MNIST(root=".",
                            train=True,
                            download=True,
                            #  A function/transform that takes in the ta
                            transform=transforms.ToTensor())

# Get the MNIST test dataset
test_data = datasets.MNIST(root=".",
                           train=False,
```

```
                                 download=True,
                                 transform=transforms.ToTensor())
```

```
train_data, test_data
```

```
(Dataset MNIST
     Number of datapoints: 60000
     Root location: .
     Split: Train
     StandardTransform
 Transform: ToTensor(),
 Dataset MNIST
     Number of datapoints: 10000
     Root location: .
     Split: Test
     StandardTransform
 Transform: ToTensor())
```

```
len(train_data), len(test_data)
```

```
(60000, 10000)
```

```
# Data is in tuple form (image, label)
img = train_data[0][0]
label = train_data[0][1]
print(f"Image:\n {img}")
print(f"Label:\n {label}")
```

```
Image:
 tensor([[[0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000],
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000],
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000],
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000],
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000],
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.000
          0.0000, 0.0000, 0.0000, 0.0000, 0.0118, 0.0706, 0.0706, 0.070
          0.4941, 0.5333, 0.6863, 0.1020, 0.6510, 1.0000, 0.9686, 0.498
          0.0000, 0.0000, 0.0000, 0.0000],
```

```
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.1176, 0.1412, 0.3686, 0.6039, 0.6667, 0.9922, 0.9922, 0.9922
            0.9922, 0.9922, 0.8824, 0.6745, 0.9922, 0.9490, 0.7647, 0.2510
            0.0000, 0.0000, 0.0000, 0.0000],
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.1922
            0.9333, 0.9922, 0.9922, 0.9922, 0.9922, 0.9922, 0.9922, 0.9922
            0.9922, 0.9843, 0.3647, 0.3216, 0.3216, 0.2196, 0.1529, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000],
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0706
            0.8588, 0.9922, 0.9922, 0.9922, 0.9922, 0.9922, 0.7765, 0.7137
            0.9686, 0.9451, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000],
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.3137, 0.6118, 0.4196, 0.9922, 0.9922, 0.8039, 0.0431, 0.0000
            0.1686, 0.6039, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000],
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0549, 0.0039, 0.6039, 0.9922, 0.3529, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000],
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.5451, 0.9922, 0.7451, 0.0078, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000],
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0431, 0.7451, 0.9922, 0.2745, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000],
           [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000, 0.1373, 0.9451, 0.8824, 0.6275
            0.4235, 0.0039, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
            0.0000, 0.0000, 0.0000, 0.0000],
```

```
# Check out the shapes of our data
print(f"Image shape: {img.shape} -> [color_channels, height, width] (CHW
print(f"Label: {label} -> no shape, due to being integer")
```

```
Image shape: torch.Size([1, 28, 28]) -> [color_channels, height, width] (
Label: 5 -> no shape, due to being integer
```

Note: There are two main agreed upon ways for representing images in machine learning:

1. Color channels first: [color_channels, height, width] (CHW) -> PyTorch default (as of April 2022)
2. Color channels last: [height, width, color_channels] (HWC) -> Matplotlib/TensorFlow default (as of April 2022)

```
# Get the class names from the dataset
class_names = train_data.classes
```

```
class_names
```

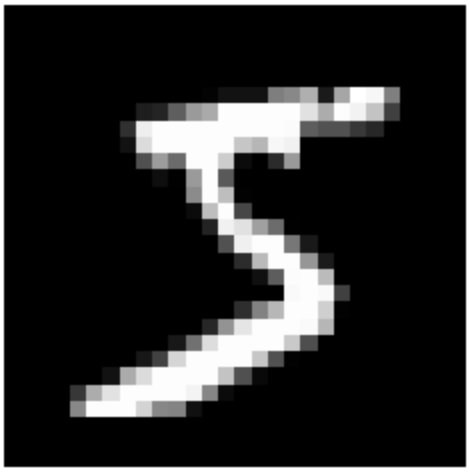
```
['0 - zero',
 '1 - one',
 '2 - two',
 '3 - three',
 '4 - four',
 '5 - five',
 '6 - six',
 '7 - seven',
 '8 - eight',
 '9 - nine']
```

6. Visualize at least 5 different samples of the MNIST training dataset.

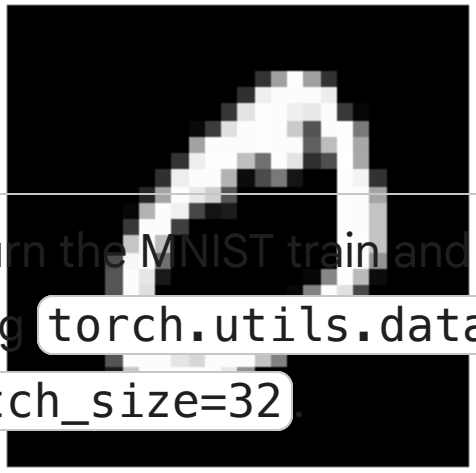
```
import matplotlib.pyplot as plt
for i in range(5):
  img = train_data[i][0]
  print(img.shape)
  img_squeeze = img.squeeze()
  print(img_squeeze.shape)
  label = train_data[i][1]
  plt.figure(figsize=(3, 3))
  plt.imshow(img_squeeze, cmap="gray")
  plt.title(label)
  plt.axis(False);
```

```
torch.Size([1, 28, 28])
torch.Size([28, 28])
torch.Size([1, 28, 28])
torch.Size([28, 28])
torch.Size([1, 28, 28])
torch.Size([28, 28])
torch.Size([1, 28, 28])
torch.Size([28, 28])
torch.Size([1, 28, 28])
torch.Size([28, 28])
```

5



0



7. Turn the MNIST train and test datasets into dataloaders using `torch.utils.data.DataLoader`, set the `batch_size=32`.

```
from torch.utils.data import DataLoader, TensorDataset
# 2. Create a DataLoader
batch_size = 32
train_dataloader = DataLoader(dataset=train_data,
                              batch_size=batch_size,
                              shuffle=True)
test_dataloader = DataLoader(dataset=test_data,
                             batch_size=batch_size,
                             shuffle=True)
```

```
train_dataloader, test_dataloader
```

```
(<torch.utils.data.dataloader.DataLoader at 0x7d1924f1ca40>,
 <torch.utils.data.dataloader.DataLoader at 0x7d19251478c0>)
```

```
for sample in next(iter(train_dataloader)):
  print(sample.shape)
```

```
torch.Size([32, 1, 28, 28])
torch.Size([32])
```

```
len(train_dataloader), len(test_dataloader)
```

```
(1875, 313)
```

8. Recreate `model_2` used in notebook 03 (the same model from the CNN Explainer website, also known as TinyVGG) capable of fitting on the MNIST dataset.

```
from torch import nn

class MNIST_model(torch.nn.Module):
  def __init__(self,input_shape: int, hidden_units: int, output_shape: i
    """
    a kernel, also known as a filter, is a small matrix of weights that
    to detect specific features like edges, textures, or shapes, produci
    stride is the number of pixels the filter (kernel) moves across the
    adds extra pixels (often zeros or reflected values) around the input
    """
    super().__init__()
    self.conv_block_1 = nn.Sequential(
      nn.Conv2d(in_channels=input_shape,
                out_channels=hidden_units,
                kernel_size=3,
                stride=1,
                padding=1),
      nn.ReLU(),
      nn.Conv2d(in_channels=hidden_units,
                out_channels=hidden_units,
                kernel_size=3,
                stride=1,
                padding=1),
      nn.ReLU(),
        # A MaxPool kernel size of 2 (typically 2x2) in Convolutional Ne
        # effectively downsampling the image, reducing computation, and
        nn.MaxPool2d(kernel_size=2)
```

```
        )
        self.conv_block_2 = nn.Sequential(
          nn.Conv2d(in_channels=hidden_units,
                    out_channels=hidden_units,
                    kernel_size=3,
                    stride=1,
                    padding=1),
          nn.ReLU(),
          nn.Conv2d(in_channels=hidden_units,
                    out_channels=hidden_units,
                    kernel_size=3,
                    stride=1,
                    padding=1),
          nn.ReLU(),
          nn.MaxPool2d(kernel_size=2)
        )
        self.classifier = nn.Sequential(
          nn.Flatten(),
          nn.Linear(in_features=hidden_units*7*7,
                    out_features=output_shape)
        )

      def forward(self, x):
        x = self.conv_block_1(x)
        # print(f"Output shape of conv block 1: {x.shape}")
        x = self.conv_block_2(x)
        # print(f"Output shape of conv block 2: {x.shape}")
        x = self.classifier(x)
        # print(f"Output shape of classifier: {x.shape}")
        return x
```

```
device = "cuda" if torch.cuda.is_available() else "cpu"
device
```

```
'cuda'
```

```
model = MNIST_model(input_shape=1,hidden_units=10,output_shape=10).to(de
model
```

```
MNIST_model(
  (conv_block_1): Sequential(
    (0): Conv2d(1, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (1): ReLU()
    (2): Conv2d(10, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
```

```
    (3): ReLU()
    (4): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (conv_block_2): Sequential(
    (0): Conv2d(10, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (1): ReLU()
    (2): Conv2d(10, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (3): ReLU()
    (4): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=490, out_features=10, bias=True)
  )
)
```

```
# Check out the model state dict to find out what patterns our model wan
#model.state_dict()
```

```
# Try a dummy forward pass to see what shapes our data is
dummy_x = torch.rand(size=(1, 28, 28)).unsqueeze(dim=0).to(device)
# dummy_x.shape
model(dummy_x)
```

```
tensor([[-0.0404,  0.0511,  0.0782,  0.0243, -0.0637,  0.0392,  0.0109,
0.0089,
         0.0672,  0.0085]], device='cuda:0', grad_fn=<AddmmBackward0>)
```

```
dummy_x_2 = torch.rand(size=([1, 10, 7, 7]))
dummy_x_2.shape
```

```
torch.Size([1, 10, 7, 7])
```

```
flatten_layer = nn.Flatten()
flatten_layer(dummy_x_2).shape
```

```
torch.Size([1, 490])
```

9. Train the model you built in exercise 8. for 5 epochs on CPU and GPU and see how long it takes on each.

```
%%time
from tqdm.auto import tqdm
```

```
# Train on CPU
model_cpu = MNIST_model(input_shape=1,
                        hidden_units=10,
                        output_shape=10).to("cpu")

# Create a loss function and optimizer
loss_fn = nn.CrossEntropyLoss()
optimizer = torch.optim.SGD(model_cpu.parameters(), lr=0.1)

### Training loop
epochs = 5
for epoch in tqdm(range(epochs)):
  train_loss = 0
  for batch, (X, y) in enumerate(train_dataloader):
    model_cpu.train()

    # Put data on CPU
    X, y = X.to("cpu"), y.to("cpu")

    # Forward pass
    y_pred = model_cpu(X)

    # Loss calculation
    loss = loss_fn(y_pred, y)
    train_loss += loss

    # Optimizer zero grad
    optimizer.zero_grad()

    # Loss backward
    loss.backward()

    # Step the optimizer
    optimizer.step()

  # Adjust train loss for number of batches
  train_loss /= len(train_dataloader)

  ### Testing loop
  test_loss_total = 0

  # Put model in eval mode
  model_cpu.eval()

  # Turn on inference mode
  with torch.inference_mode():
    for batch, (X_test, y_test) in enumerate(test_dataloader):
      # Make sure test data on CPU
      X_test, y_test = X_test.to("cpu"), y_test.to("cpu")
      test_pred = model_cpu(X_test)
      test_loss = loss_fn(test_pred, y_test)
```

```
      test_loss_total += test_loss

    test_loss_total /= len(test_dataloader)

  # Print out what's happening
  print(f"Epoch: {epoch} | Loss: {train_loss:.3f} | Test loss: {test_los
```

```
100%                                                 5/5 [03:00<00:00, 36.05s/it]
Epoch: 0 | Loss: 0.301 | Test loss: 0.082
Epoch: 1 | Loss: 0.082 | Test loss: 0.068
Epoch: 2 | Loss: 0.063 | Test loss: 0.053
Epoch: 3 | Loss: 0.054 | Test loss: 0.043
Epoch: 4 | Loss: 0.047 | Test loss: 0.046
CPU times: user 2min 56s, sys: 1 s, total: 2min 57s
Wall time: 3min
```

```
%%time
from tqdm.auto import tqdm

device = "cuda" if torch.cuda.is_available() else "cpu"

# Train on GPU
model_gpu = MNIST_model(input_shape=1,
                        hidden_units=10,
                        output_shape=10).to(device)

# Create a loss function and optimizer
loss_fn = nn.CrossEntropyLoss()
optimizer = torch.optim.SGD(model_gpu.parameters(), lr=0.1)

# Training loop
epochs = 5
for epoch in tqdm(range(epochs)):
  train_loss = 0
  model_gpu.train()
  for batch, (X, y) in enumerate(train_dataloader):
    # Put data on target device
    X, y = X.to(device), y.to(device)

    # Forward pass
    y_pred = model_gpu(X)

    # Loss calculation
    loss = loss_fn(y_pred, y)
    train_loss += loss

    # Optimizer zero grad
    optimizer.zero_grad()
```

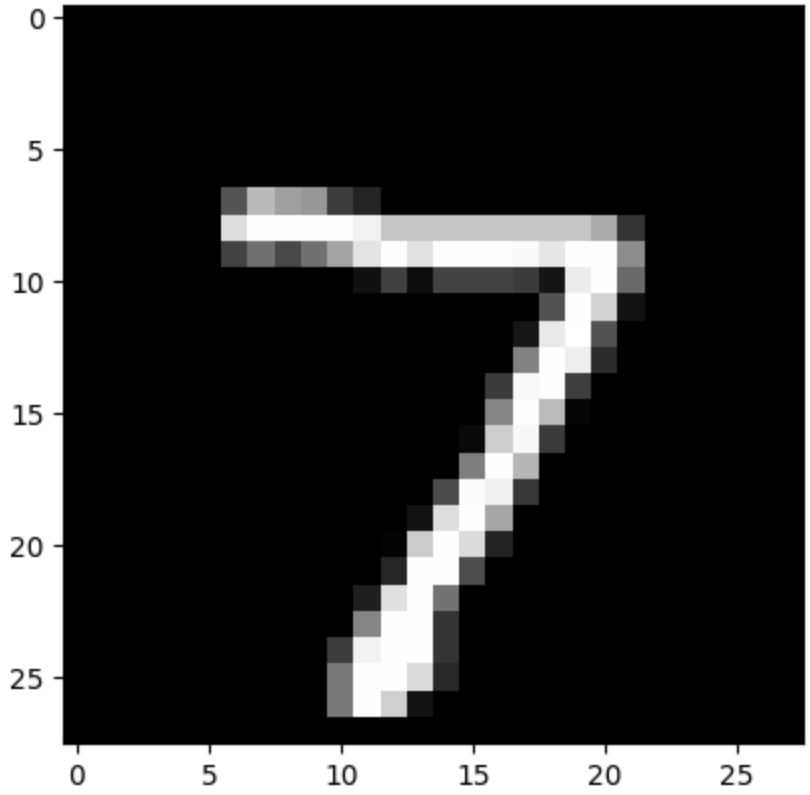
```
    # Loss backward
    loss.backward()

    # Step the optimizer
    optimizer.step()

  # Adjust train loss to number of batches
  train_loss /= len(train_dataloader)

  ### Testing loop
  test_loss_total = 0
  # Put model in eval mode and turn on inference mode
  model_gpu.eval()
  with torch.inference_mode():
    for batch, (X_test, y_test) in enumerate(test_dataloader):
      # Make sure test data on target device
      X_test, y_test = X_test.to(device), y_test.to(device)

      test_pred = model_gpu(X_test)
      test_loss = loss_fn(test_pred, y_test)

      test_loss_total += test_loss

    # Adjust test loss total for number of batches
    test_loss_total /= len(test_dataloader)

  # Print out what's happening
  print(f"Epoch: {epoch} | Loss: {train_loss:.3f} | Test loss: {test_los
```

```
100%                                                 5/5 [00:58<00:00, 11.53s/it]
Epoch: 0 | Loss: 0.423 | Test loss: 0.081
Epoch: 1 | Loss: 0.080 | Test loss: 0.055
Epoch: 2 | Loss: 0.063 | Test loss: 0.060
Epoch: 3 | Loss: 0.054 | Test loss: 0.056
Epoch: 4 | Loss: 0.048 | Test loss: 0.040
CPU times: user 57 s, sys: 411 ms, total: 57.4 s
Wall time: 58 s
```

10. Make predictions using your trained model and visualize at least 5 of them comparing the prediciton to the target label.

```
plt.imshow(test_data[0][0].squeeze(),cmap="gray")
```

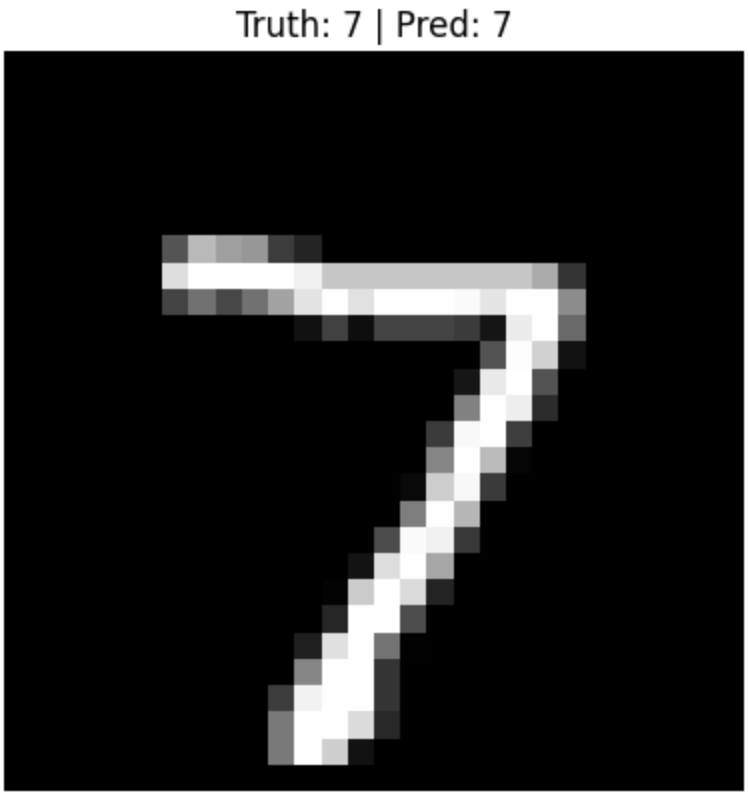
```
<matplotlib.image.AxesImage at 0x7d1924b0b590>
```

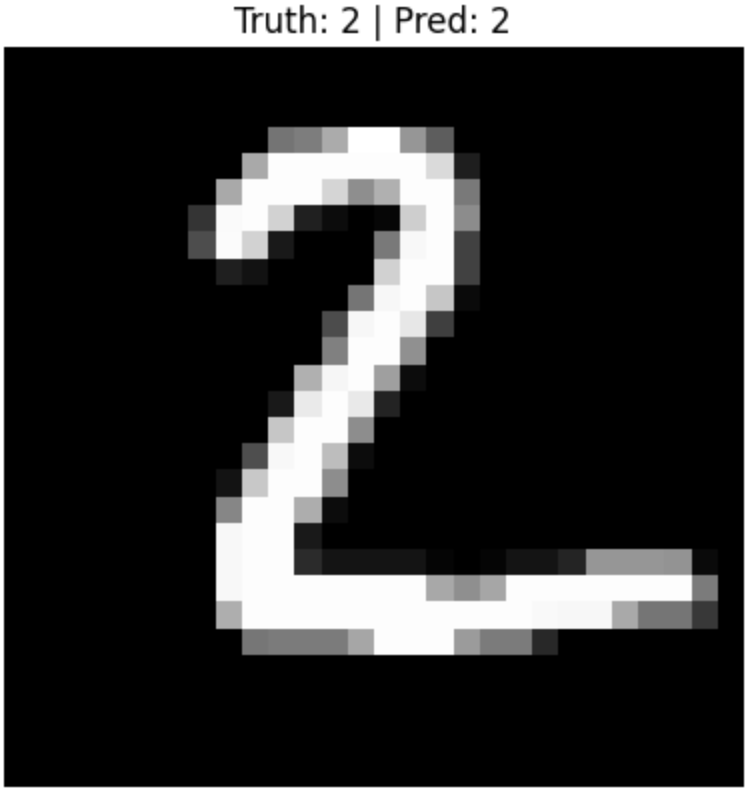


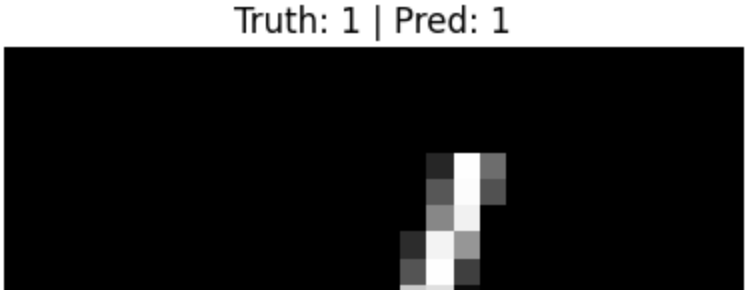

```
# Logits -> Prediction probabilities -> Prediction labels
model_pred_logits = model_gpu(test_data[0][0].unsqueeze(dim=0).to(device
model_pred_probs = torch.softmax(model_pred_logits, dim=1)
model_pred_label = torch.argmax(model_pred_probs, dim=1)
model_pred_label
```

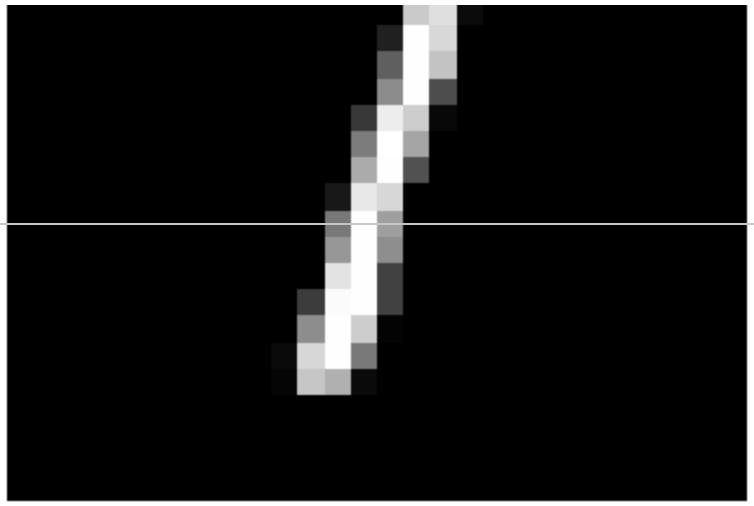
```
tensor([7], device='cuda:0')
```

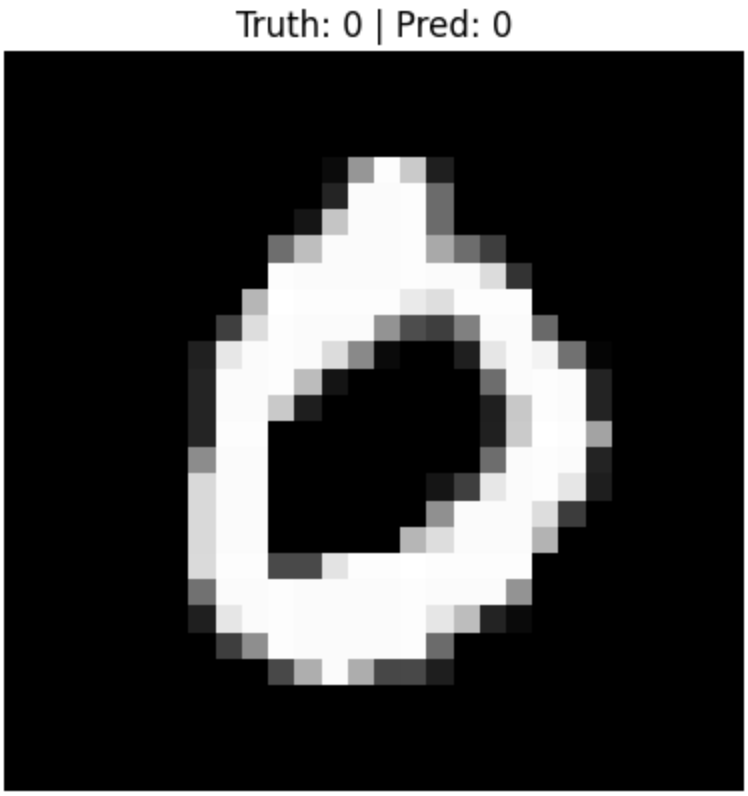
```
num_to_plot = 5
for i in range(num_to_plot):
  # Get image and labels from the test data
  img = test_data[i][0]
  label = test_data[i][1]

  # Make prediction on image
  model_pred_logits = model_gpu(img.unsqueeze(dim=0).to(device))
  model_pred_probs = torch.softmax(model_pred_logits, dim=1)
  model_pred_label = torch.argmax(model_pred_probs, dim=1)

  # Plot the image and prediction
  plt.figure()
  plt.imshow(img.squeeze(), cmap="gray")
  plt.title(f"Truth: {label} | Pred: {model_pred_label.cpu().item()}")
  plt.axis(False);
```

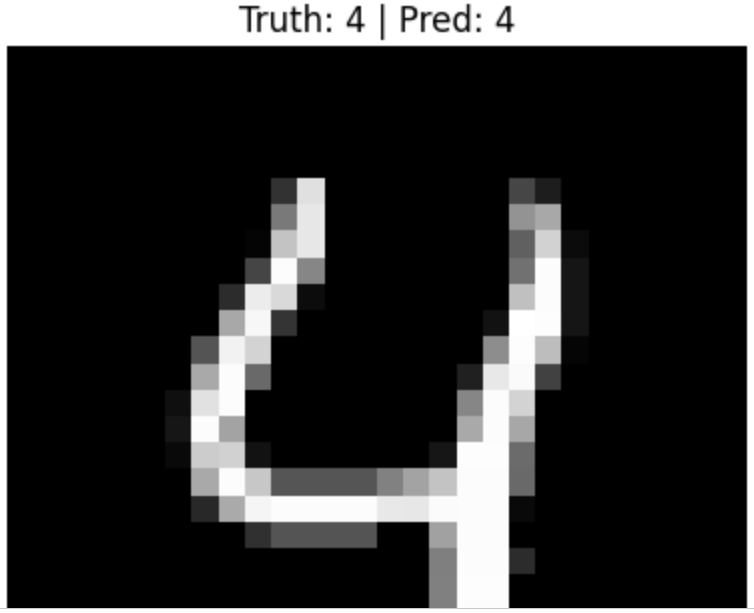

Truth: 7 | Pred: 7


Truth: 2 | Pred: 2


Truth: 1 | Pred: 1




Truth: 0 | Pred: 0


Truth: 4 | Pred: 4

11. Plot a confusion matrix comparing your model's predictions to the truth labels.

```
# See if torchmetrics exists, if not, install it
try:
    import torchmetrics, mlxtend
    print(f"mlxtend version: {mlxtend.__version__}")
    assert int(mlxtend.__version__.split(".")[1]) >= 19, "mlxtend veriso
except:
    !pip install -q torchmetrics -U mlxtend # <- Note: If you're using G
    import torchmetrics, mlxtend
    print(f"mlxtend version: {mlxtend.__version__}")
```

```
mlxtend version: 0.24.0
```

```
# Import mlxtend upgraded version
import mlxtend
print(mlxtend.__version__)
assert int(mlxtend.__version__.split(".")[1]) >= 19 # should be version
```

```
0.24.0
```

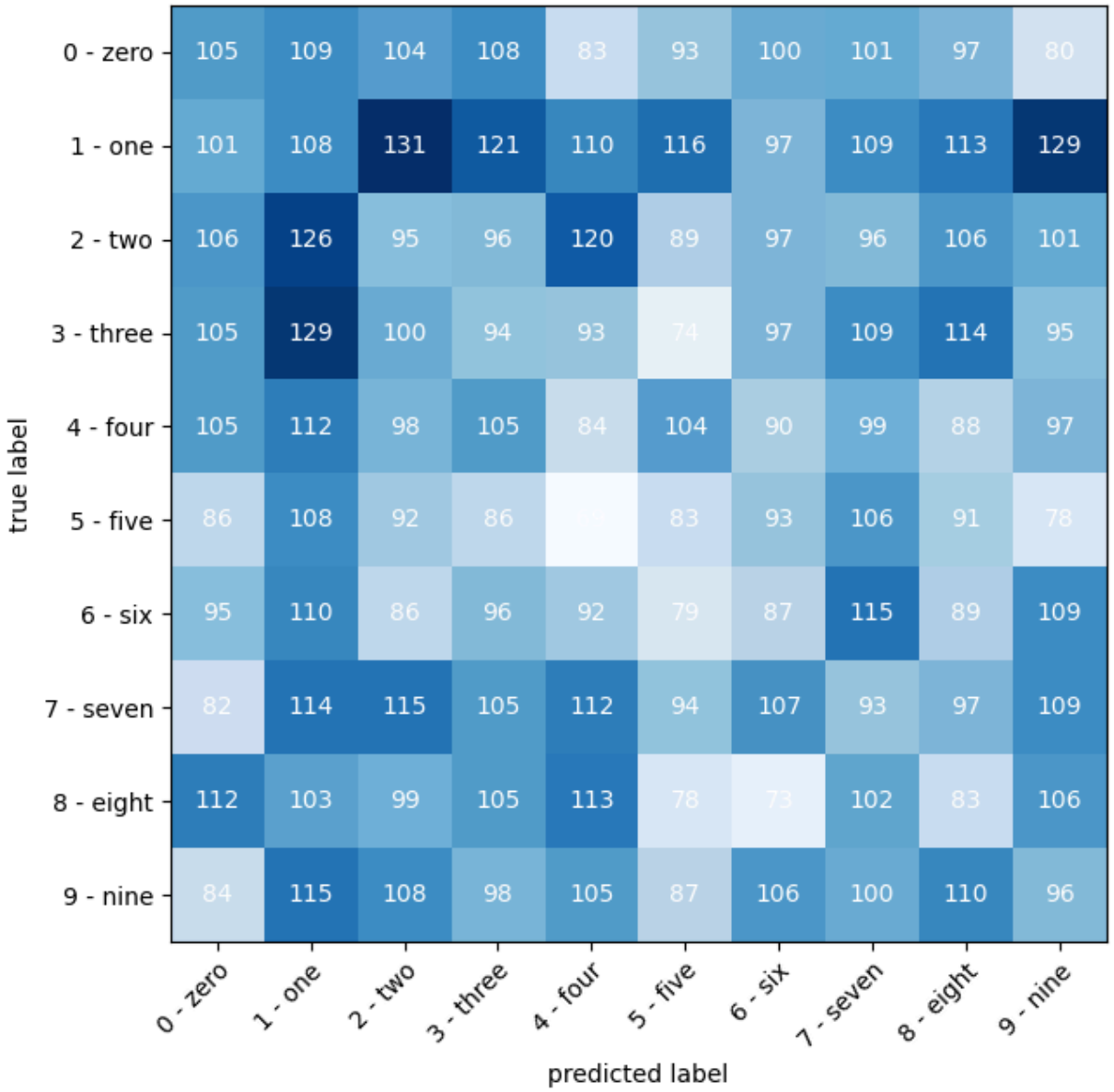
```
# Make predictions across all test data
from tqdm.auto import tqdm
model_gpu.eval()
y_preds = []
with torch.inference_mode():
  for batch, (X, y) in tqdm(enumerate(test_dataloader)):
    # Make sure data on right device
    X, y = X.to(device), y.to(device)
    # Forward pass
    y_pred_logits = model_gpu(X)
    # Logits -> Pred probs -> Pred label
    y_pred_labels = torch.argmax(torch.softmax(y_pred_logits, dim=1), di
    # Append the labels to the preds list
    y_preds.append(y_pred_labels)
  y_preds=torch.cat(y_preds).cpu()
len(y_preds)
```

```
313/? [00:01<00:00, 270.58it/s]
10000
```

```
test_data.targets[:10], y_preds[:10]
```

```
(tensor([7, 2, 1, 0, 4, 1, 4, 9, 5, 9]),
 tensor([9, 0, 5, 4, 0, 6, 9, 8, 7, 3]))
```

```
from torchmetrics import ConfusionMatrix
from mlxtend.plotting import plot_confusion_matrix

# Setup confusion matrix
confmat = ConfusionMatrix(task="multiclass", num_classes=len(class_names
confmat_tensor = confmat(preds=y_preds,
                         target=test_data.targets)

# Plot the confusion matrix
fix, ax = plot_confusion_matrix(
    conf_mat=confmat_tensor.numpy(),
    class_names=class_names,
    figsize=(10, 7)
)
```



12. Create a random tensor of shape `[1, 3, 64, 64]` and pass it through a `nn.Conv2d()` layer with various hyperparameter settings

```
random_tensor = torch.rand([1, 3, 64, 64])
random_tensor.shape
```

```
torch.Size([1, 3, 64, 64])
```

```
conv_layer = nn.Conv2d(in_channels=3,
                       out_channels=64,
                       kernel_size=3,
                       stride=2,
                       padding=1)

print(f"Random tensor original shape: {random_tensor.shape}")
random_tensor_through_conv_layer = conv_layer(random_tensor)
print(f"Random tensor through conv layer shape: {random_tensor_through_c
```

```
Random tensor original shape: torch.Size([1, 3, 64, 64])
Random tensor through conv layer shape: torch.Size([1, 64, 32, 32])
```

13. Use a model similar to the trained `model_2` from notebook 03 to make predictions on the test `torchvision.datasets.FashionMNIST` dataset.

- Then plot some predictions where the model was wrong alongside what the label of the image should've been.
- After visualing these predictions do you think it's more of a modelling error or a data error?
- As in, could the model do better or are the labels of the data too close to each other (e.g. a "Shirt" label is too close to "T-shirt/top")?

```
# Download FashionMNIST train & test
from torchvision import datasets
from torchvision import transforms

fashion_mnist_train = datasets.FashionMNIST(root=".",
                                            download=True,
                                            train=True,
                                            transform=transforms.ToTenso
```

```
fashion_mnist_test = datasets.FashionMNIST(root=".",
                                           train=False,
                                           download=True,
                                           transform=transforms.ToTensor

len(fashion_mnist_train), len(fashion_mnist_test)
```

```
100%|██████████| 26.4M/26.4M [00:02<00:00, 11.8MB/s]
100%|██████████| 29.5k/29.5k [00:00<00:00, 189kB/s]
100%|██████████| 4.42M/4.42M [00:01<00:00, 3.11MB/s]
100%|██████████| 5.15k/5.15k [00:00<00:00, 25.2MB/s]
(60000, 10000)
```

```
# Get the class names of the Fashion MNIST dataset
fashion_mnist_class_names = fashion_mnist_train.classes
fashion_mnist_class_names
```

```
['T-shirt/top',
 'Trouser',
 'Pullover',
 'Dress',
 'Coat',
 'Sandal',
 'Shirt',
 'Sneaker',
 'Bag',
 'Ankle boot']
```

```
# Turn FashionMNIST datasets into dataloaders
from torch.utils.data import DataLoader

fashion_mnist_train_dataloader = DataLoader(fashion_mnist_train,
                                            batch_size=32,
                                            shuffle=True)

fashion_mnist_test_dataloader = DataLoader(fashion_mnist_test,
                                           batch_size=32,
                                           shuffle=False)

len(fashion_mnist_train_dataloader), len(fashion_mnist_test_dataloader)
```

```
(1875, 313)
```

```
# model_2 is the same architecture as MNIST_model
model_2 = MNIST_model(input_shape=1,
                      hidden_units=10,
                      output_shape=10).to(device)
model_2
```

```
MNIST_model(
  (conv_block_1): Sequential(
    (0): Conv2d(1, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (1): ReLU()
    (2): Conv2d(10, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (3): ReLU()
    (4): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (conv_block_2): Sequential(
    (0): Conv2d(10, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (1): ReLU()
    (2): Conv2d(10, 10, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (3): ReLU()
    (4): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=490, out_features=10, bias=True)
  )
)
```

```
# Setup loss and optimizer
from torch import nn
loss_fn = nn.CrossEntropyLoss()
optimizer = torch.optim.SGD(model_2.parameters(), lr=0.01)
```

```
# Setup metrics
from tqdm.auto import tqdm
from torchmetrics import Accuracy

acc_fn = Accuracy(task = 'multiclass', num_classes=len(fashion_mnist_cla

# Setup training/testing loop
epochs = 1000
for epoch in tqdm(range(epochs)):
  train_loss, test_loss_total = 0, 0
  train_acc, test_acc = 0, 0

  ### Training
  model_2.train()
  for batch, (X_train, y_train) in enumerate(fashion_mnist_train_dataloa
    X_train, y_train = X_train.to(device), y_train.to(device)

    # Forward pass and loss
    y_pred = model_2(X_train)
    loss = loss_fn(y_pred, y_train)
```